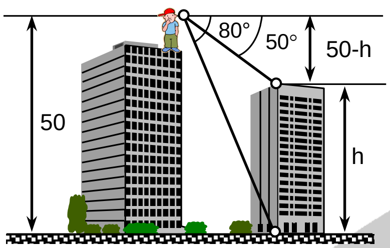
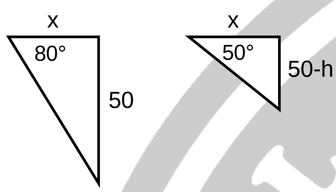
Solutions:

1. A. 39.49



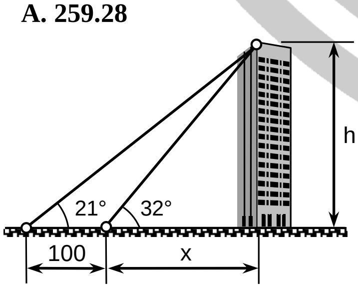


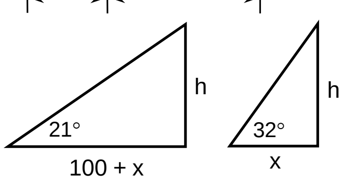
$$\tan 80 = \frac{50}{x}$$
$$x = 8.816 \text{ m}$$
$$\tan 50^\circ = \frac{50 - h}{x}$$
$$\tan 50^\circ = \frac{50 - h}{8.816}$$
$$\tan 50^\circ (8.816) = 50 - h$$
$$10.506 = 50 - h$$
$$h = 39.49 \text{ m}$$

2. A. – cos A

$$\sin(270^\circ - A) = \begin{bmatrix} \sin 270^\circ \cos A \\ - \sin A \cos 270^\circ \end{bmatrix}$$
$$\sin(270^\circ - A) = (-1)\cos A - \sin A(0)$$
$$\sin(270^\circ - A) = -\cos A$$

3. A. 259.28





$$\tan 21^\circ = \frac{h}{100 + x}$$
$$h = (100 + x)\tan 21^\circ \rightarrow \text{Eq.1}$$
$$\tan 32^\circ = \frac{h}{x}$$
$$h = x \tan 32^\circ \rightarrow \text{Eq.2}$$

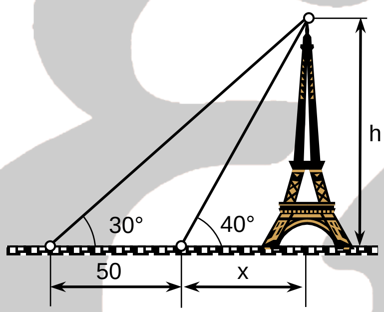
Equating Eq.1 to Eq.2:

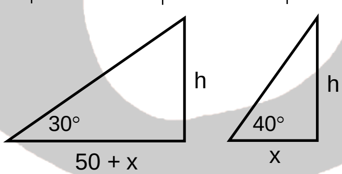
$$(100 + x)\tan 21^\circ = x \tan 32^\circ$$
$$100 + x = x\left(\frac{\tan 32^\circ}{\tan 21^\circ}\right)$$
$$100 + x = 1.6278x$$
$$x = 159.286 \text{ m}$$

Thus:

$$x + 100 = 259.286 \text{ m}$$

4. B. 92.54 feet



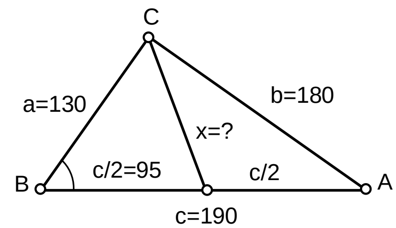


$$\tan 40^\circ = \frac{h}{x}$$
$$x = \frac{h}{\tan 40^\circ} \rightarrow \text{Eq.1}$$
$$\tan 30^\circ = \frac{h}{50 + x}$$
$$x = \frac{h}{\tan 30^\circ} - 50 \rightarrow \text{Eq.2}$$

Equate Eq.1 to Eq.2:

$$\frac{h}{\tan 40°} = \frac{h}{\tan 30°} - 50$$
$$1.19175h = 1.73205h - 50$$
$$h = 92.54 \text{ ft.}$$

5. C. 125 m



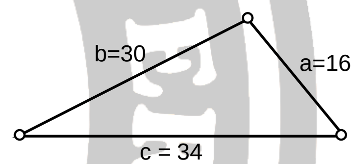
By cosine law:

$$b^2 = a^2 + c^2 - 2ac \cos B$$
$$180^2 = 130^2 + 190^2 - 2(130)(190)\cos B$$
$$B = 65.35^\circ$$

By cosine law:

$$x^2 = a^2 + (c/2)^2 - 2a(c/2)\cos B$$
$$x^2 = 130^2 + 95^2 - 2(130)(95)\cos 65.35^\circ$$
$$x = 125 \text{ m}$$

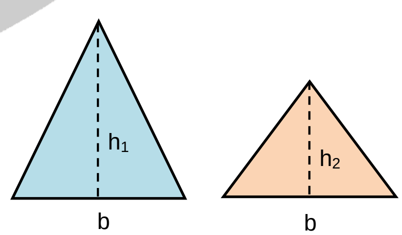
6. A. 240



$$S = \frac{a + b + c}{2}$$
$$S = \frac{16 + 30 + 34}{2}$$
$$S = 40$$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$
$$A = \sqrt{40(40-16)(40-30)(40-34)}$$
$$A = 240 \text{ square units}$$

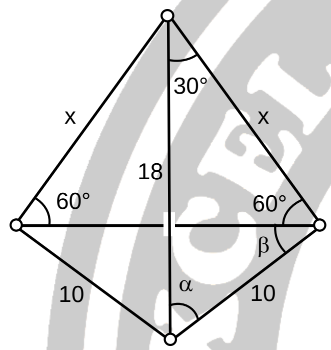
7. D. 4 and 10



$$h_1 = b + 3$$
$$h_2 = b - 3$$

$$A_1 = A_2 + 21$$
$$\frac{1}{2}bh_1 = \frac{1}{2}bh_2 + 21$$
$$b(b+3) = b(b-3) + 42$$
$$\cancel{b^2} + 3b = \cancel{b^2} - 3b + 42$$
$$6b = 42$$
$$b = 7$$
$$h_1 = 7 + 3 = 10 \text{ units}$$
$$h_2 = 7 - 3 = 4 \text{ units}$$

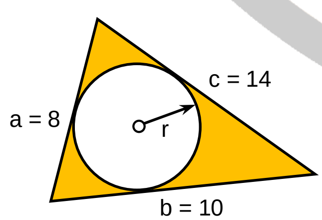
8. C. 19.95 m



By sine law:
$$\frac{\sin 30}{10} = \frac{\sin(60+\beta)}{18}$$
$$\beta = 4.158^\circ$$
$$\alpha + \beta + 60 + 30 = 180$$
$$\alpha + 4.158 + 90 = 180$$
$$\alpha = 85.842^\circ$$

By sine law:
$$\frac{\sin 85.842^\circ}{x} = \frac{\sin 30^\circ}{10}$$
$$x = 19.95 \text{ m}$$

9. C. 2.45 cm



$$S = \frac{a+b+c}{2}$$
$$S = \frac{8+10+14}{2}$$
$$S = 16 \text{ cm}$$

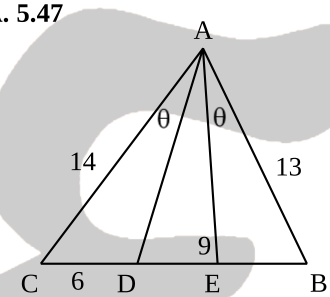
$$A = \sqrt{s(s-a)(s-b)(s-c)}$$
$$A = \sqrt{16(16-8)(16-10)(16-14)}$$
$$A = 39.19 \text{ cm}^2$$

$$A = rs$$
$$39.19 = r(16)$$
$$r = 2.45 \text{ cm}$$

10. D. incenter

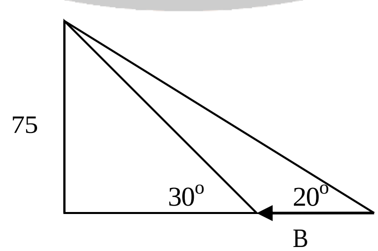
11. B. hypotenuse/2

12. A. 5.47



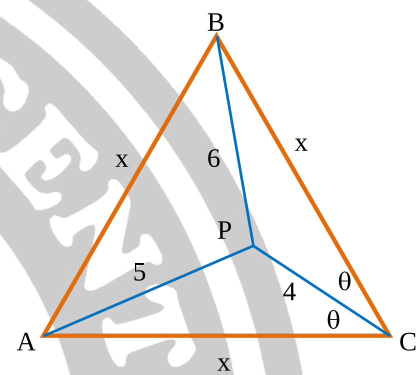
$$\frac{BE}{\sin\theta} = \frac{13}{\sin AEB}$$
$$\frac{9}{\sin BAD} = \frac{13}{\sin AEB}$$
$$\frac{BE}{9} = \frac{\sin\theta}{\sin BAD}$$
$$\frac{6}{\sin\theta} = \frac{14}{\sin ACD}$$
$$\frac{15-BE}{\sin CAE} = \frac{14}{\sin ACD}$$
$$\frac{15-BE}{14} = \frac{\sin\theta}{\sin CAE}$$
$$\frac{BE}{9} = \frac{15-BE}{14}$$
$$BE = 5.87$$

13. C. 4.57



$$d = \frac{75}{\tan(20^\circ)} - \frac{75}{\tan(30^\circ)}$$
$$d = 76.16$$
$$v = \frac{d}{t}$$
$$v = \frac{76.16 \text{ m}}{1 \text{ min}}\left(\frac{1 \text{ km}}{1000\text{m}}\right)\left(\frac{60 \text{ min}}{1 \text{ hr}}\right)$$
$$v = 4.57 \text{ kph}$$

14. C. 8.53 units



In ΔBCP, using cosine law:
$$4^2 + x^2 - 2(4)x\cos\theta = 6^2$$
$$8x\cos\theta = x^2 - 20 \rightarrow \text{Eq.1}$$

In ΔACP, using cosine law:
$$4^2 + x^2 - 2(4)x\cos(60-\theta) = 5^2$$
$$8x\cos(60-\theta) = x^2 - 9$$
$$8x[\cos 60\cos\theta + \sin 60\sin\theta] = x^2 - 9$$
$$4x\cos\theta + 6.9282x\sin\theta = x^2 - 9 \rightarrow \text{Eq.2}$$

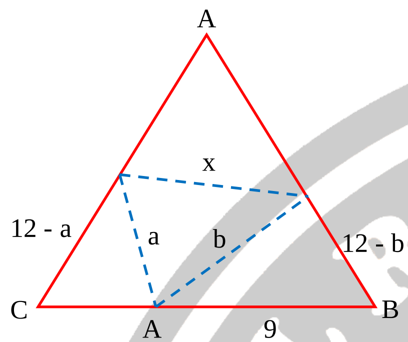
From Eq. 1:
$$x\cos\theta = \frac{x^2-20}{8}$$

Then we have:
$$\left[\begin{array}{l} 4\left(\frac{1}{8}\right)(x^2-20) \\ +6.9282x\sin\theta \end{array}\right] = x^2 - 9$$
$$6.9282x\sin\theta = 0.5x^2 + 1$$
$$\sin^2\theta = \frac{0.25x^4 + x^2 + 1}{48x^2}$$

From Eq. 1:
$$\cos\theta = \frac{x^2-20}{8x}$$
$$\cos^2\theta = \frac{x^4 - 40x^2 + 400}{64x^2}$$
$$\sin^2\theta + \cos^2\theta = 1$$

$$\frac{0.25x^4 - x^2 + 1}{48x^2} + \frac{x^4 - 40x^2 + 400}{64x^2} = 1$$
$$x = 8.53$$

15. B. 6.96



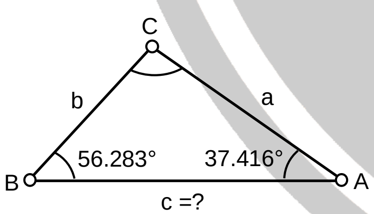
$$\angle A = \angle B = \angle C = 60^o$$

$$a^2 = (12-a)^2 + 9^2 - 2(9)(12-a)\cos 60^o$$
$$a = 4.2$$

$$b^2 = (12-a)^2 + 3^2 - 2(3)(12-a)\cos 60^o$$
$$b = 6.43$$

$$x^2 = a^2 + b^2 - 2ab\cos 60^o$$
$$x = \sqrt{4.2^2 + 6.43^2 - 2(4.2)(6.43)\cos 60^o}$$
$$x = 6.96$$

16. B. 181.5 m



A = 37°25' = 37.416°
B = 56°17' = 56.283°

$$A + B + C = 180$$
$$37.416 + 56.283 + C = 180$$
$$C = 86.301^\circ$$

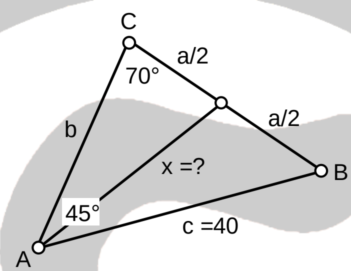
By sine law:

$$\frac{\sin 86.301°}{c} = \frac{\sin 37.416°}{a}$$

$$a = \left(\frac{\sin 86.301}{\sin 37.416}\right)c$$
$$a = 0.609\,c$$

$$A_{\text{triangle}} = \frac{1}{2}ac\sin B$$
$$8346 = \frac{1}{2}(0.609c)(c)\sin 56.283)$$
$$16{,}692 = 0.5065\,c^2$$
$$c = 181.5$$

17. A. 36.3 m



By sine law:

$$\frac{\sin C}{c} = \frac{\sin A}{a}$$
$$\frac{\sin 70°}{40} = \frac{\sin 45°}{a}$$
$$a = 30\ \text{m}$$
$$\frac{a}{2} = 15\ \text{m}$$
$$A + B + C = 180$$
$$45 + B + 70 = 180$$
$$B = 65^\circ$$

By cosine law:

$$x = \sqrt{c^2 + (a/2)^2 - 2(c)(a/2)\cos B}$$
$$x = \sqrt{(40)^2 + (15)^2 - 2(40)(15)\cos 65^\circ}$$
$$x = 36.3\,\text{m}$$

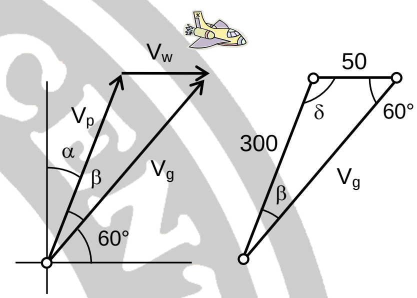
18. A. 0

$$x = \frac{\cos A + \cos B}{\sin A - \sin B} + \frac{\sin A + \sin B}{\cos A - \cos B}$$

$$x = \frac{\begin{pmatrix}(\cos A + \cos B)(\cos A - \cos B) \\ + (\sin A + \sin B)(\sin A - \sin B)\end{pmatrix}}{(\sin A - \sin B)(\cos A - \cos B)}$$

$$x = \frac{(\cos^2 A - \cos^2 B) + (\sin^2 A - \sin^2 B)}{(\sin A - \sin B)(\cos A - \cos B)}$$
$$x = \frac{(\cos^2 A + \sin^2 A) - (\sin^2 B + \cos^2 B)}{(\sin A - \sin B)(\cos A - \cos B)}$$
$$x = \frac{1 - 1}{(\sin A - \sin B)(\cos A - \cos B)}$$
$$x = 0$$

19. C. 21.7°, 321.8 mph



By sine law:

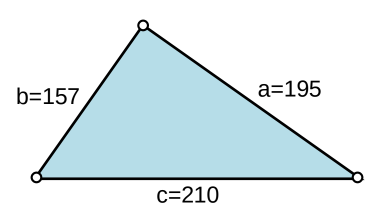
$$\frac{50}{\sin\beta} = \frac{300}{\sin 60}$$
$$\beta = 8.3^\circ$$

$$\alpha + \beta + 60 = 180$$
$$\alpha + 8.3 + 60 = 180$$
$$\alpha = 111.7^\circ$$

By sine law:

$$\frac{V}{\sin 111.7} = \frac{50}{\sin 8.3}$$
$$V = \frac{\sin 111.7}{\sin .3}(50)$$
$$V = 321.8\ \text{mph}$$

20. C. 14,586 sq. units



$$S = \frac{a+b+c}{2}$$

$$S = \frac{195+157+210}{2}$$

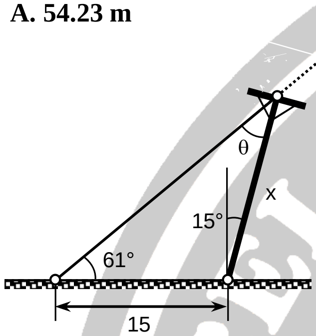
$$S = 281$$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$A = \sqrt{281(281-195)(281-157)(281-210)}$$

$$A = 14{,}586.2 \text{ square units}$$

21. A. 54.23 m



$$\theta + 61 + 90 + 15 = 180$$

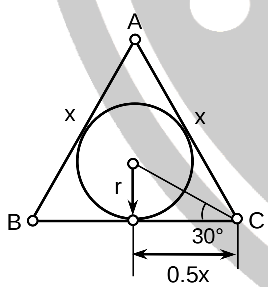
$$\theta = 14^\circ$$

By sine law:

$$\frac{\sin 14^\circ}{15} = \frac{\sin 61^\circ}{x}$$

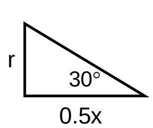
$$x = 54.23\,\text{m}$$

22. A. 34.64 cm



Since equilateral triangle, A = B = C = 60°

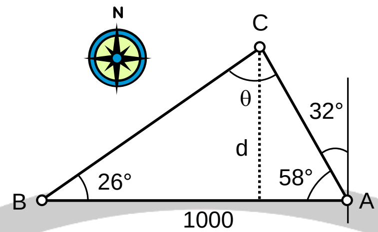
$$\tan 30^\circ = \frac{r}{0.5x}$$



$$\tan 30^\circ = \frac{10}{0.5x}$$

$$x = 34.64\,\text{cm}$$

23. B. 374 m



$$\theta + 26 + 58 = 180$$

$$\theta = 96^\circ$$

By sine law:

$$\frac{\sin 96^\circ}{1000} = \frac{\sin 58^\circ}{BC}$$

$$BC = 852.719\,\text{m}$$

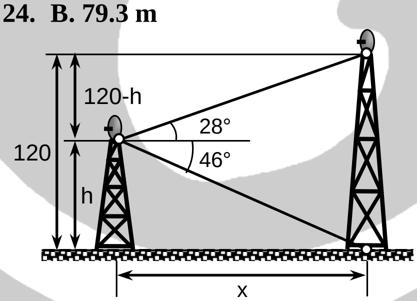
$$\sin 26^\circ = \frac{d}{BC}$$

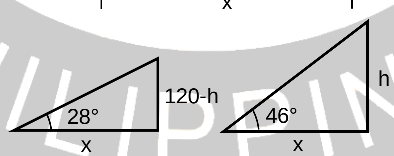
$$\sin 26^\circ = \frac{d}{852.719}$$

$$d = 852.719(\sin 26^\circ)$$

$$d = 374\,\text{m}$$

24. B. 79.3 m





$$\tan 28^\circ = \frac{120-h}{x}$$

$$x = \frac{120-h}{\tan 28^\circ} \rightarrow \text{Eq.1}$$

$$\tan 46^\circ = \frac{h}{x}$$

$$x = \frac{h}{\tan 46^\circ} \rightarrow \text{Eq.2}$$

Equating Eq.1 to Eq.2:

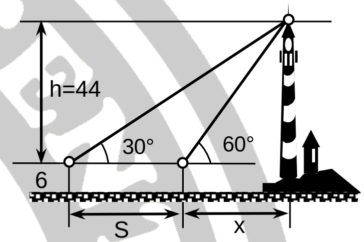
$$\frac{120-h}{\tan 28^\circ} = \frac{h}{\tan 46^\circ}$$

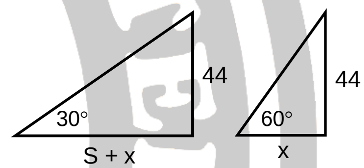
$$120 - h = \left(\frac{\tan 28^\circ}{\tan 46^\circ}\right)h$$

$$120 - h = 0.513h$$

$$h = 79.31\,\text{m}$$

25. C. 0.169 m/sec





$$\tan 60 = \frac{44}{x}$$

$$x = 25.4\,\text{m}$$

$$\tan 30 = \frac{44}{S + 25.4}$$

$$S + 25.4 = 76.21$$

$$S = 50.81\,\text{m}$$

$$V = \frac{S}{t}$$

$$V = \frac{50.81}{5(60)}$$

$$V = 0.169\,\text{m/s}$$